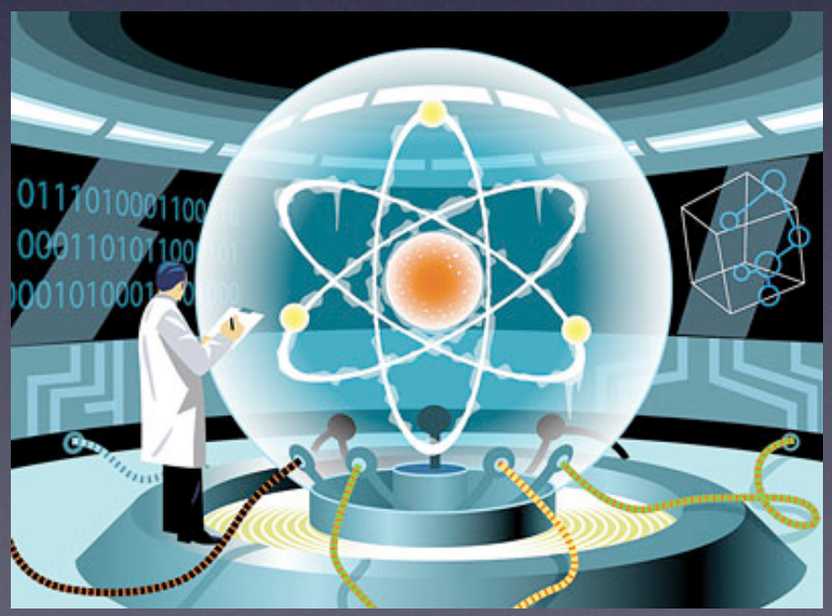
Quantum Computing Myth, Fact, and Potential

Dave Bacon
Department of Computer Science & Engineering
Department of Physics
University of Washington



Funded by

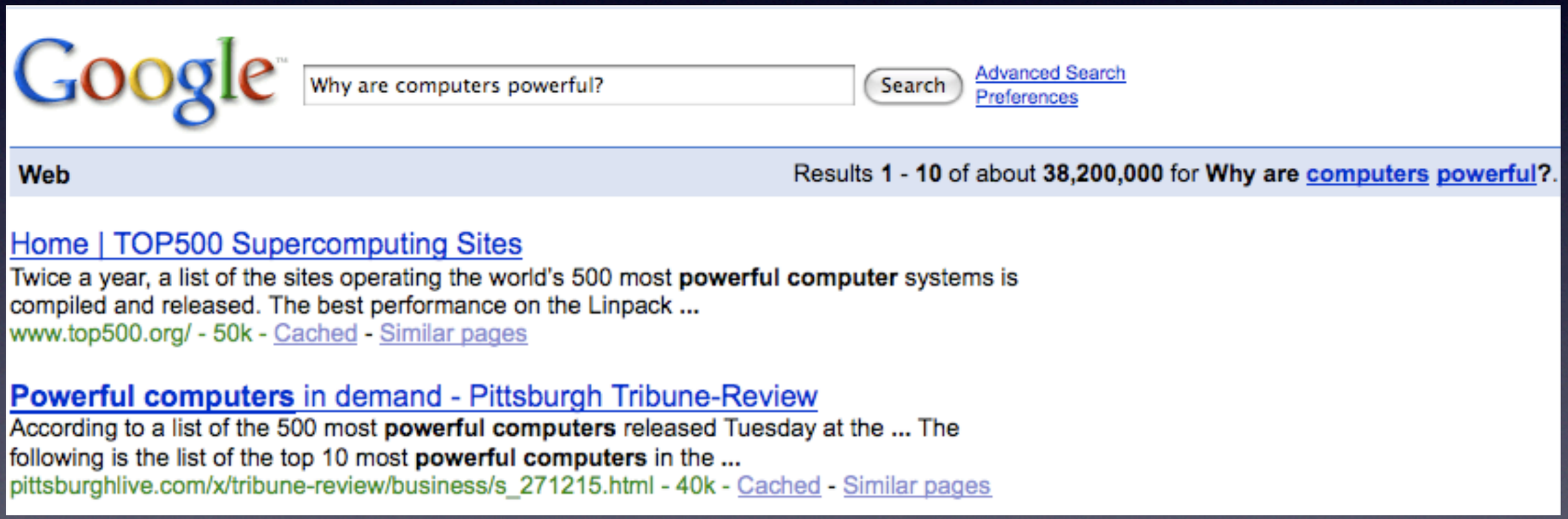
NSF ARO NSA

A Question

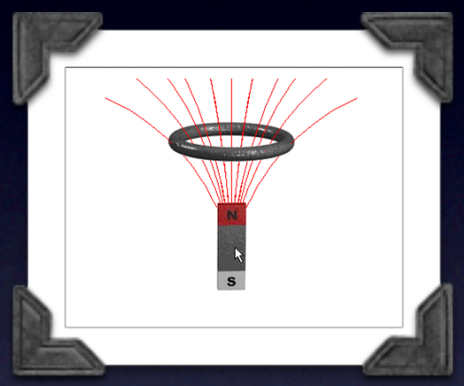
Where does the power of computers come from?

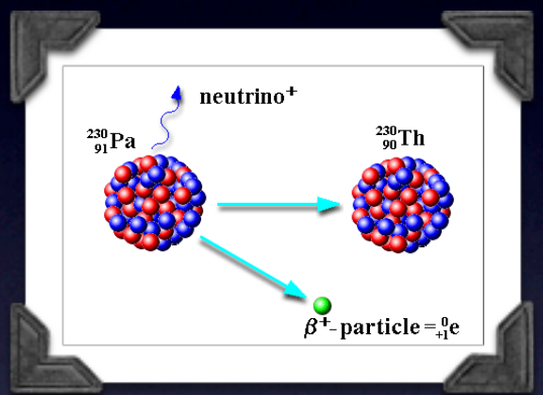
A Question

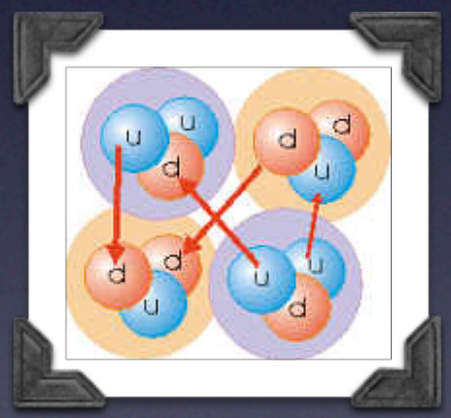
Where does the power of computers come from?

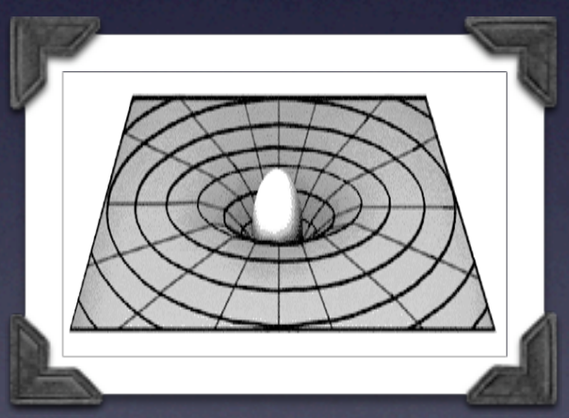


Can We Derive the Power of the Algorithm?









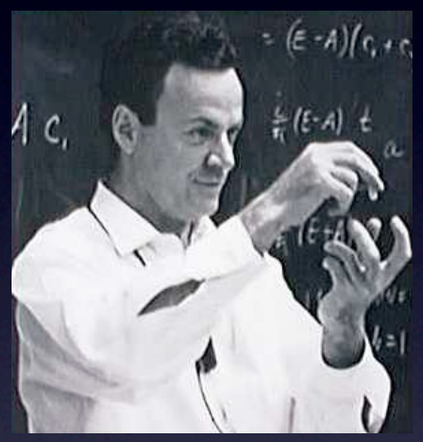
```
function extended_gcd(a, b)
    x := 0     lastx := 1
    y := 1     lasty := 0
    while b ≠ 0
        temp := b
        quotient := a div b
        b := a mod b
        a := temp

        temp := x
        x := lastx-quotient*x
        lastx := temp

        temp := y
        y := lasty-quotient*y
        lasty := temp
    return {lastx, lasty, a}
```

Physics → Computer Science

Some Simple Insights of Geniuses

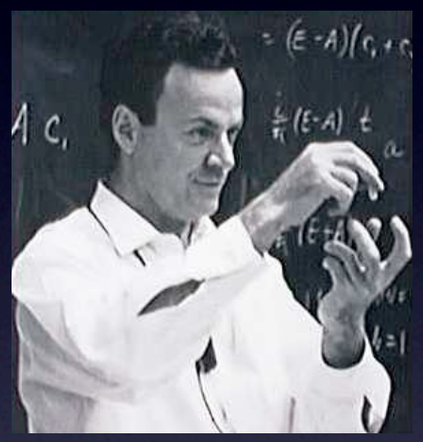


Richard Feynman



Dave Deutsch

Some Simple Insights of Geniuses



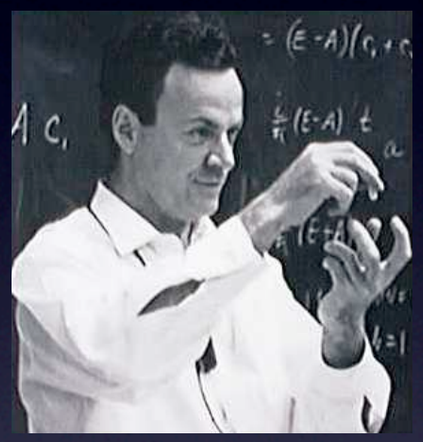
Richard Feynman



Dave Deutsch

1. Laws of physics are quantum mechanical

Some Simple Insights of Geniuses



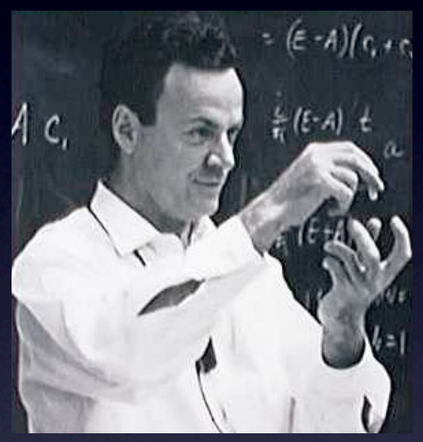
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Dave Deutsch

1. Laws of physics are quantum mechanical
2. Quantum theory appears to be hard to simulate on today's computers

Some Simple Insights of Geniuses



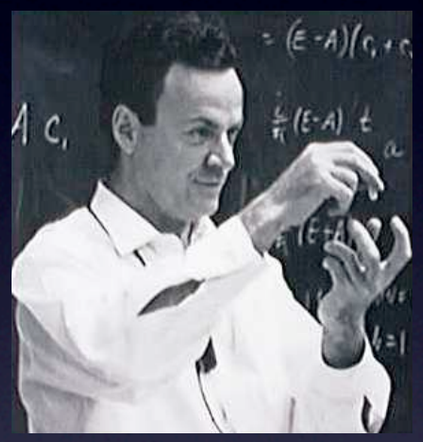
Richard Feynman



Dave Deutsch

1. Laws of physics are quantum mechanical
2. Quantum theory appears to be hard to simulate on today's computers
3. Yet somehow nature "simulates"

Some Simple Insights of Geniuses



Richard Feynman



Dave Deutsch

1. Laws of physics are quantum mechanical
2. Quantum theory appears to be hard to simulate on today's computers
3. Yet somehow nature "simulates"

Computers built from parts which obey quantum theory?

Quantum Computers!

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570599

=

x

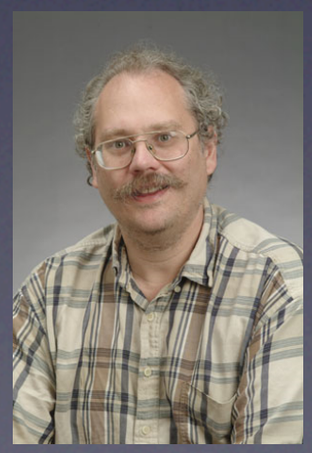
Quantum Computers!

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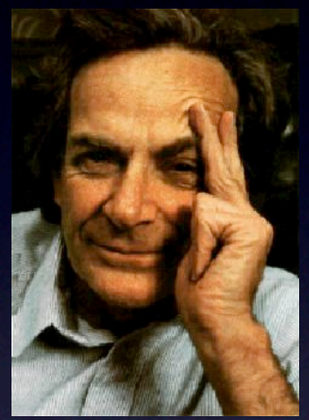
x



Peter Shor (1994)

- Quantum computers can efficiently factor
- No known classical efficient algorithm.
- Breaks RSA public key cryptosystem

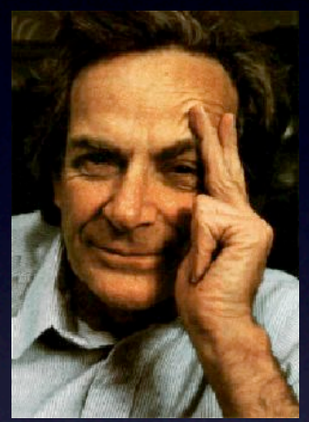
Quantum Propoganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Quantum Propoganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965



Anyone who is not shocked by quantum theory has not understood it.

Niels Bohr
Nobel Prize 1922

Hogwash



- They understand it
- They use it
- It just defies their sensibilities

Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	0 1	0 1	
Observation:			
Evolution:			

Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	$\boxed{0}$ $\boxed{1}$	$\boxed{0}$ $\boxed{1}$ $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	
Observation:			
Evolution:			

Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	$\boxed{0}$ $\boxed{1}$	$\boxed{0}$ $\boxed{1}$ $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:			
Evolution:			

Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	$\boxed{0}$ $\boxed{1}$	$\boxed{0}$ $\boxed{1}$ $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	$\boxed{0}$ certainty	$\boxed{0}$ *p* percent $\boxed{1}$ *1-p* percent	
Evolution:			

Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	0 1	0 1 $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent 1 *1-p* percent	0 $\lvert\alpha\rvert^2$percent 1 $\lvert\beta\rvert^2$percent
Evolution:			

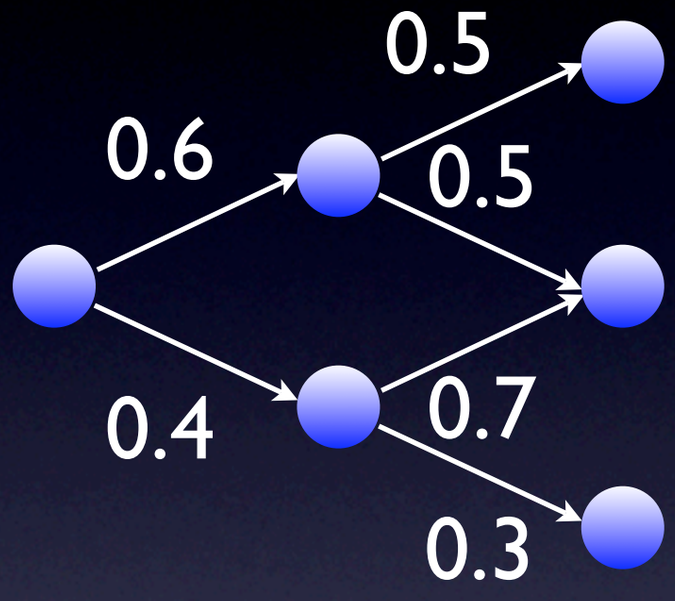
Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	$\boxed{0}$ $\boxed{1}$	$\boxed{0}$ $\boxed{1}$ $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	$\boxed{0}$ certainty	$\boxed{0}$ *p* percent $\boxed{1}$ *1-p* percent	$\boxed{0}$ $\lvert\alpha\rvert^2$percent $\boxed{1}$ $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ deterministic	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$ stochastic	

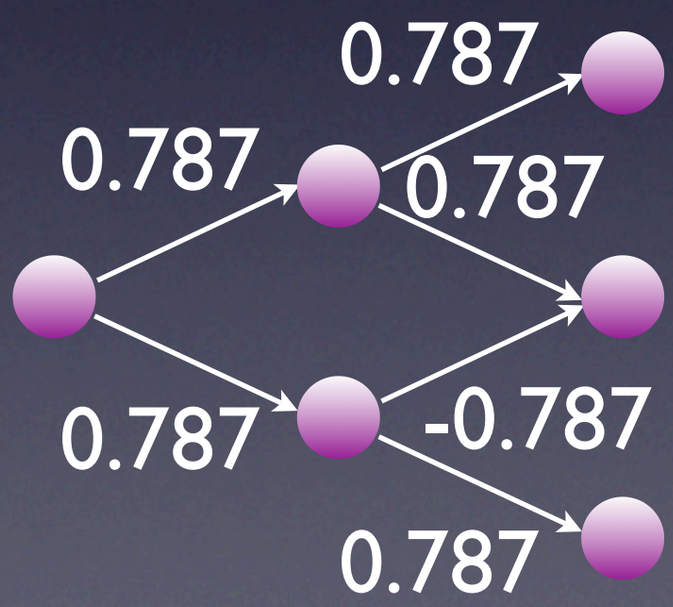
Quantum Theory.... Quickly

	bits	pbits	qubits
Description:	$\boxed{0}$ $\boxed{1}$	$\boxed{0}$ $\boxed{1}$ $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	$\boxed{0}$ certainty	$\boxed{0}$ *p* percent $\boxed{1}$ *1-p* percent	$\boxed{0}$ $\lvert\alpha\rvert^2$percent $\boxed{1}$ $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic	unitary

Garden of Re-forking Paths



$(0.6)(0.5)+(0.4)(0.7)=0.58$



$|(0.787)(0.787)+(-0.787)\ (0.787)|^2=0$

Deutsch's Problem



David Deutsch

Dr. Falcon





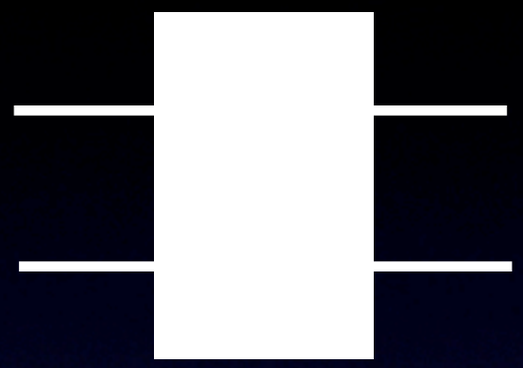
$xy \to uv$	$xy \to uv$	$xy \to uv$	$xy \to uv$
$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$

constant

balanced

Deutsch's Problem

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$



$00 \to 00$
$01 \to 01$
$10 \to 10$
$11 \to 11$

$00 \to 01$
$01 \to 00$
$10 \to 11$
$11 \to 10$

$00 \to 01$
$01 \to 00$
$10 \to 10$
$11 \to 11$

$00 \to 00$
$01 \to 01$
$10 \to 11$
$11 \to 10$

constant

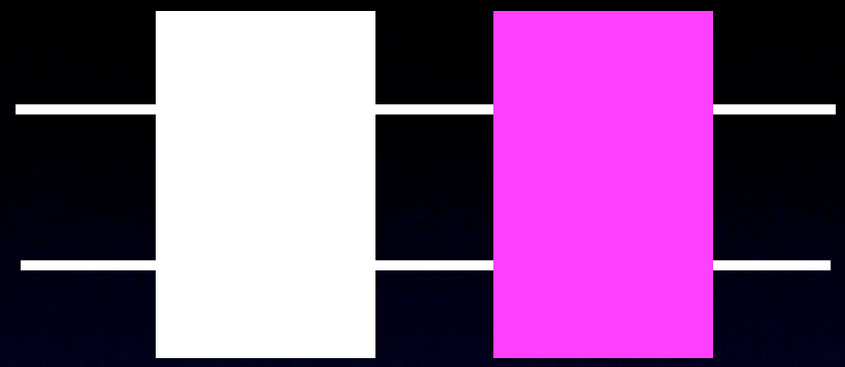
$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \qquad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

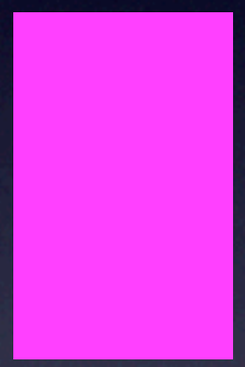
balanced

$$\begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \qquad \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Deutsch's Problem

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$





$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

constant

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

01

balanced

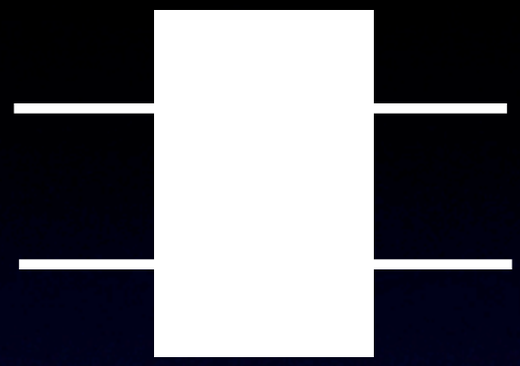
$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

11

Deutsch's Problem



1985



- Classically: 2 queries
- Quatumly: 1 query

Might the computational complexity depend on physics?

Beyond a Curiosity



Quantum computers offer superpolynomial query complexity speedups over classical computers [Bernstein, Vazirani 1993]

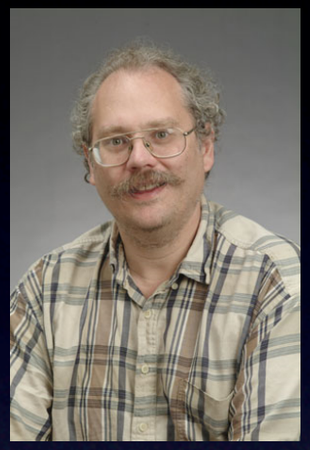
Umesh Vazirani
(Berkeley)

Quantum computers offer exponential query complexity speedups over classical computers [Simon 1994]



Dan Simon
(Microsoft)

Shor's Breakthrough



Peter Shor

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570599807508642406493739712550055038649119906436234252670840638518957594638895726176858331794727721461074353025362307197304822463291469530209711645985217113052071125636359039752747727214610743530253623071973048224632914695302097116459852171130520711256363590397527

=

Best classical algorithm	Shor's quantum algorithm
$O\left(\exp(n^{\frac{1}{3}})\right)$	$O\left(n^3\right)$

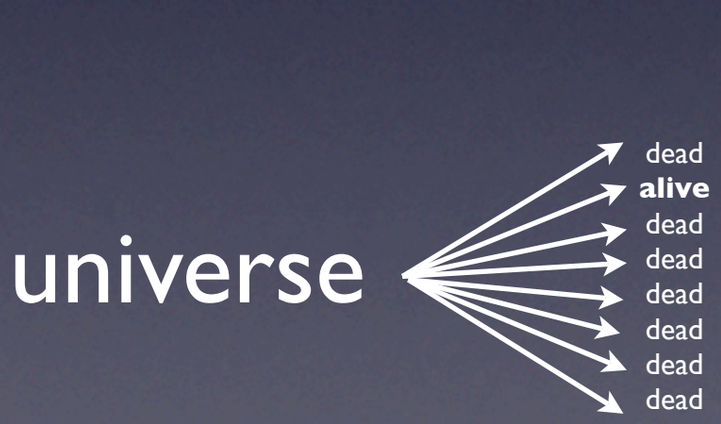
Myth, Fact, and Potential

- Myth
- Fact
- Potential

Myth: Parallelism

Claim: Quantum computers are massively parallel computers

To search over an exponentially large space, simply branch the universe into an exponential number of copies





Exponentially small probability of alive

Fact: The Power of Quantum Computers

Quantum computers are faster at some problems:

Exponential speedups: Factoring, discrete logarithm, solving Pell's equation...

Polynomial speedups: Searching for a need in a haystack, NAND tree problem, ...

Quantum computers are **not** thought to allow efficient solution to **NP-complete** problems

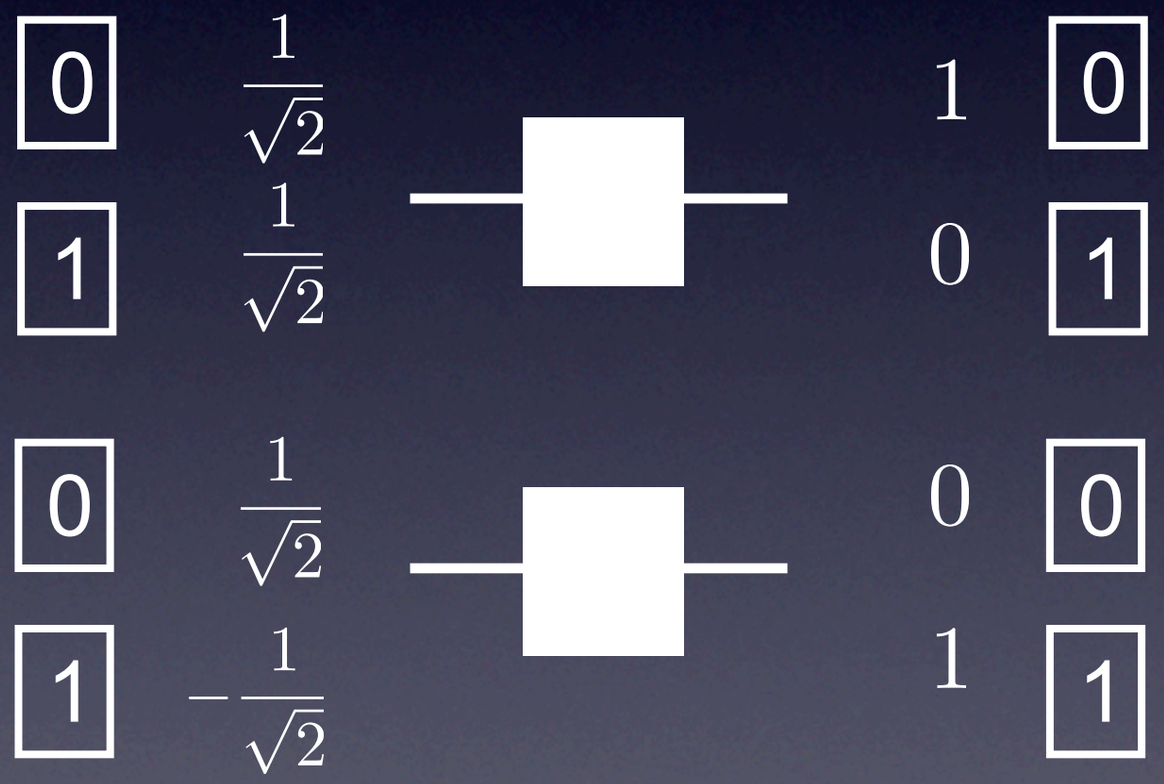
Fact: Origin of the Power of Quantum Computers

Classical

$|0\rangle$ 50 %

$|1\rangle$ 50 %

Quantum



Symmetries of quantum wave functions can be exploited

Fact: Origin of the Power of Quantum Computers

Classical correlation

0	0	50 %
1	1	50 %

Quantum entanglement

0	0	$\frac{1}{\sqrt{2}}$
1	1	$\frac{1}{\sqrt{2}}$

Bells Theorem: Entangled quantum systems exhibit correlations which cannot be explained by classical local hidden variable theories

Translation: Entanglement is genuinely quantum

Potential: Quantum Insights

"Small" Entanglement = Efficiently classical simulation

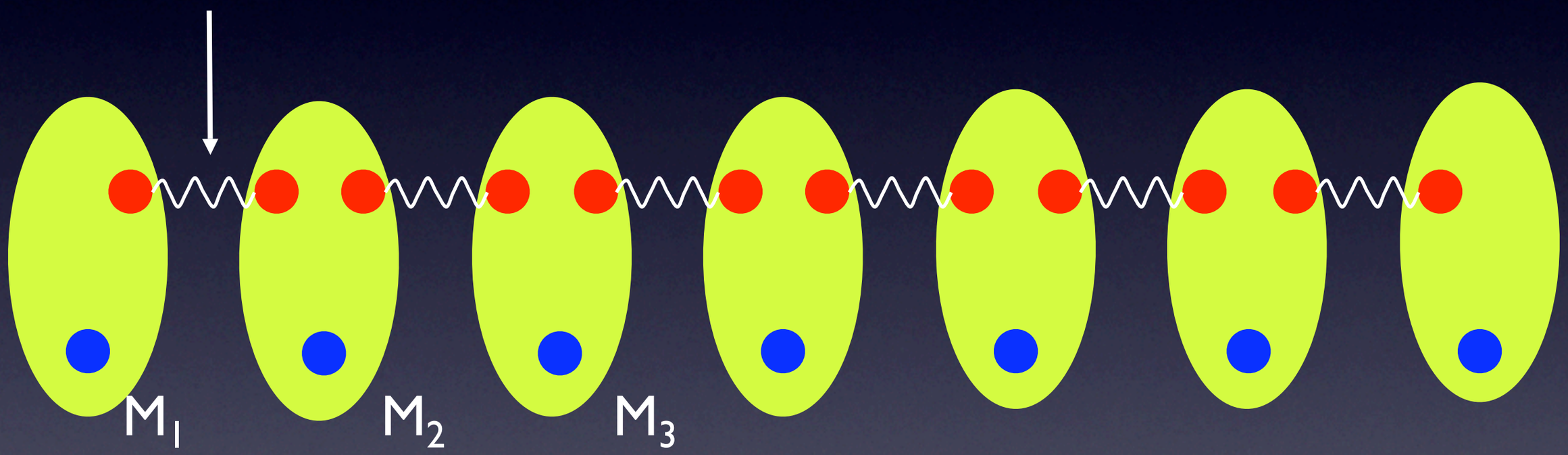
Potential: Quantum Insights

"Small" Entanglement = Efficiently classical simulation



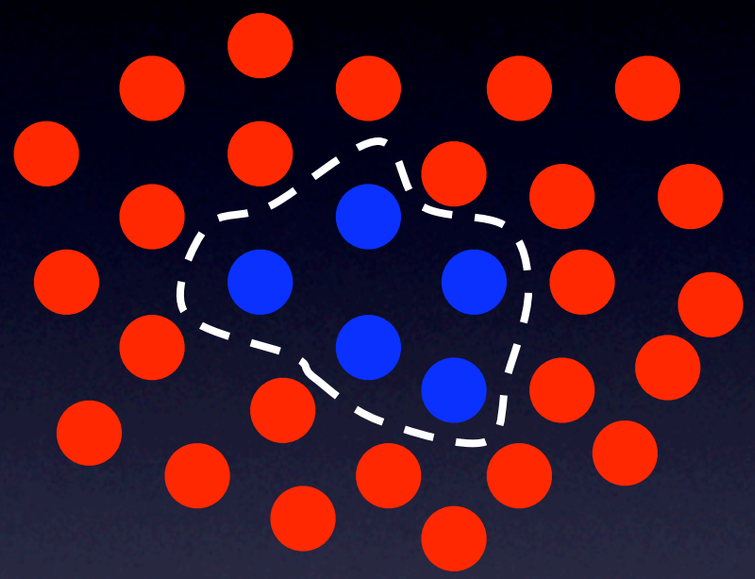
Potential: Quantum Insights

"Small" Entanglement = Efficiently classical simulation



Density Matrix Renormalization Group [White, 1991]

Potential: Quantum Insights



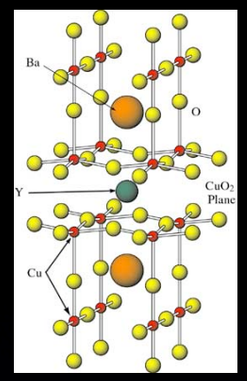
In "many" quantum systems, the low energy sector has an entropy of a region which grows like the perimeter of the region

"Small" entanglement

Design algorithms whose variational ansatz respect this scaling [Vidal, Verstraete...]

New classical algorithms for efficiently simulating quantum systems

Potential: Quantum Algorithms



Simulated Quantum Computation of Molecular Energies

Alán Aspuru-Guzik,[1]*† Anthony D. Dutoi,[1]* Peter J. Love,[2] Martin Head-Gordon[1,3]

The calculation time for the energy of atoms and molecules scales exponentially with system size on a classical computer but polynomially using quantum algorithms. We demonstrate that such algorithms can be applied to problems of chemical interest using modest numbers of quantum bits. Calculations of the water and lithium hydride molecular ground-state energies have been carried out on a quantum computer simulator using a recursive phase-estimation algorithm. The recursive algorithm reduces the number of quantum bits required for the readout register from about 20 to 4. Mappings of the molecular wave function to the quantum bits are described. An adiabatic method for the preparation of a good approximate ground-state wave function is described and demonstrated for a stretched hydrogen molecule. The number of quantum bits required scales linearly with the number of basis functions, and the number of gates required grows polynomially with the number of quantum bits.

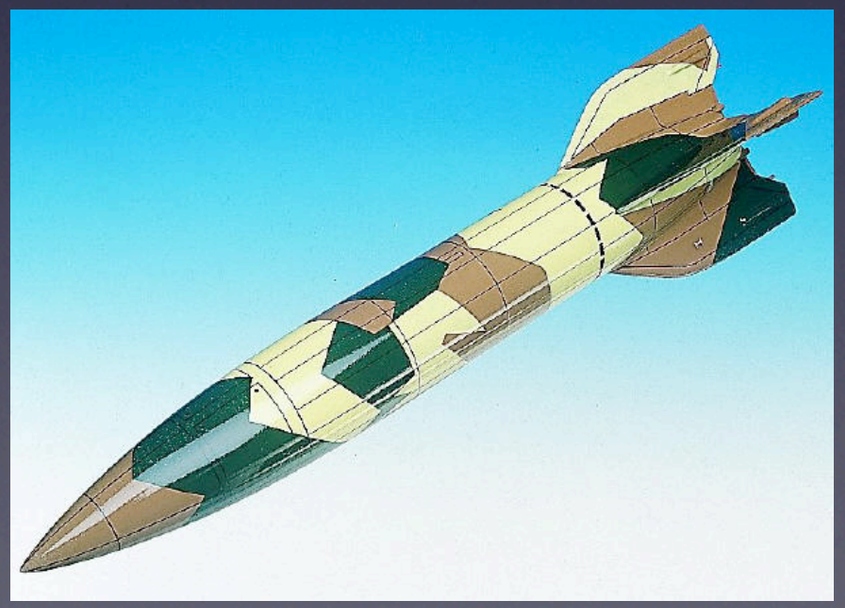
Annie Get Your Gun Principle

~50 to 100 qubit can compete with today's physical chemistry algorithms

Myth: Quantum Computers are Analog Computers

Myth: Quantum computers are analog computers

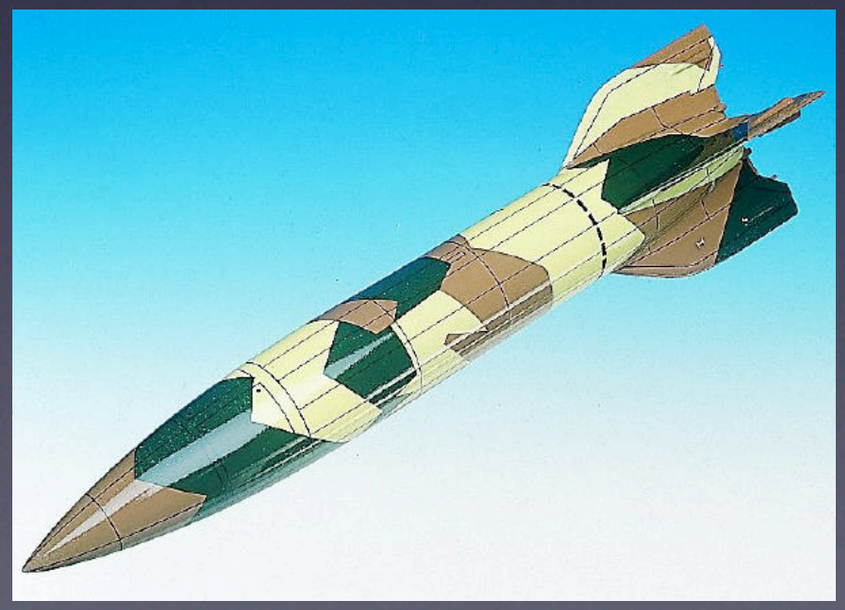
Infinite precision required to specify quantum state



Myth: Quantum Computers are Analog Computers

Myth: Quantum computers are analog computers

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \leftarrow$$ Infinite precision required to specify quantum state



Category Errors

Analog / Digital



vs



0.9982131123....

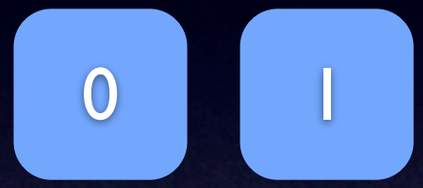
0

Category Errors

Analog / Digital



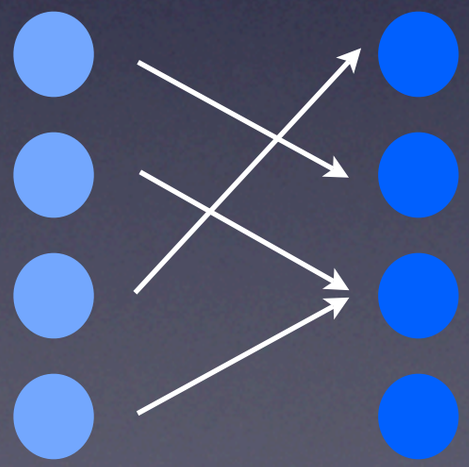
vs



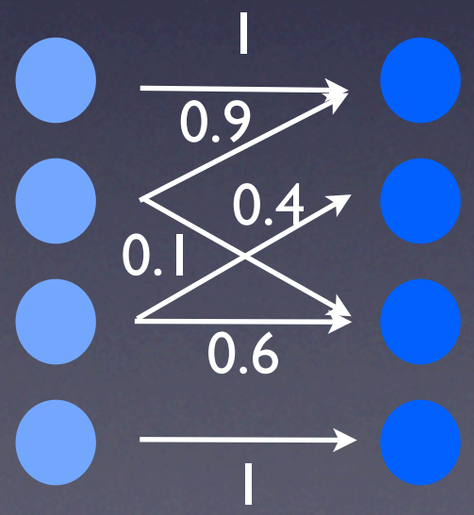
0.998213123....

0

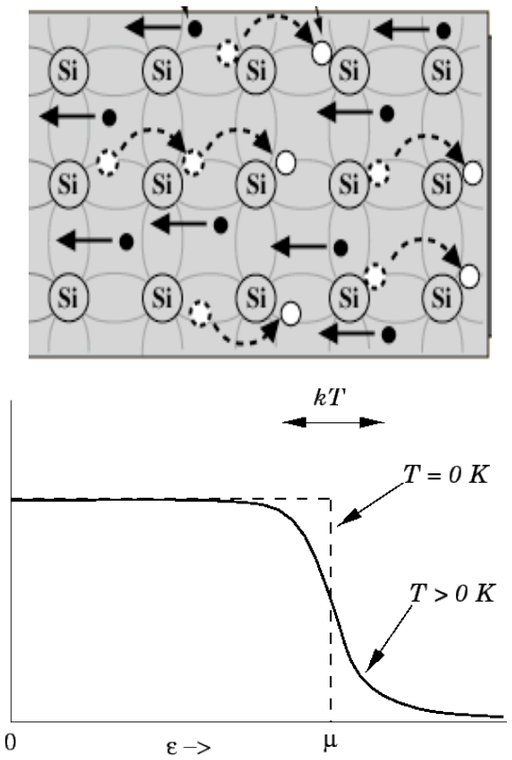
Probabilistic / Deterministic



vs



Analog Probabilistic
Digital Deterministic



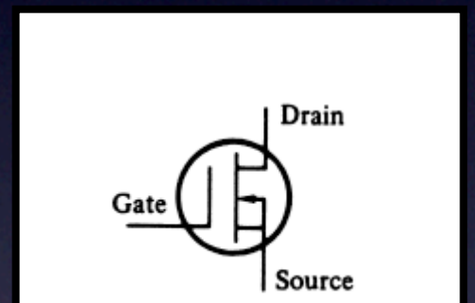
behind the bit

Analog
Voltages

Probabilistic
Electrons

Digital
Signal

Deterministic
Gates



bits

The Real Categories

Analog / Digital



vs



0.9982131 23....

0

The Real Categories

Analog / Digital



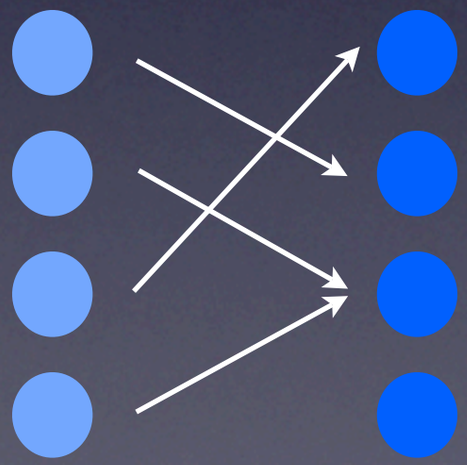
0.998213123....

vs

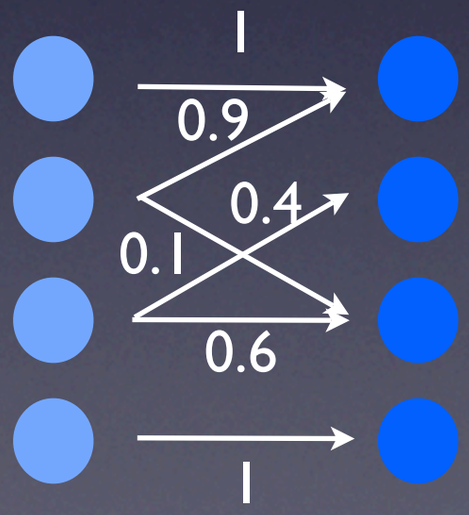


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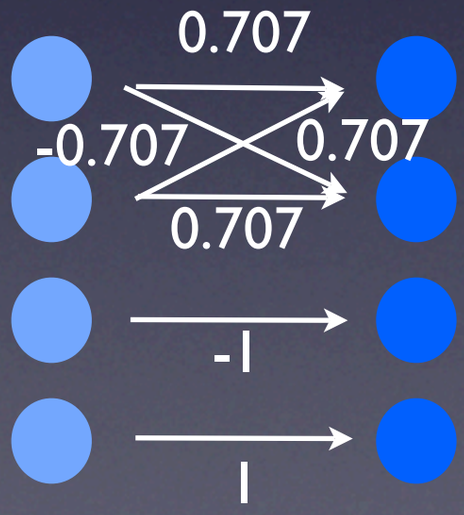
Probabilistic / Deterministic / Quantum



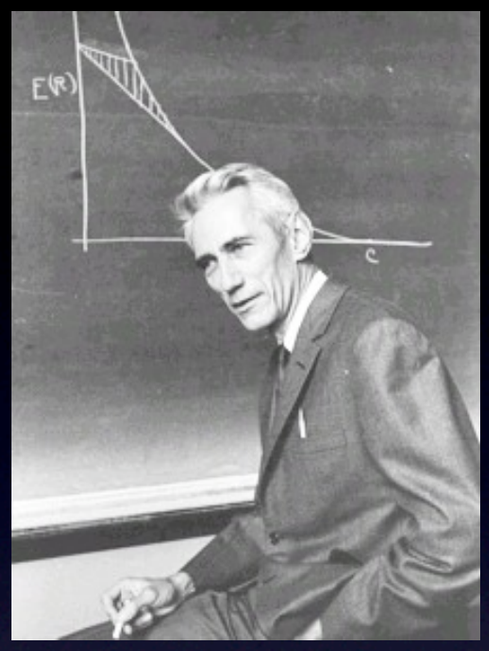
vs



vs



Noisy Probabilistic Bits



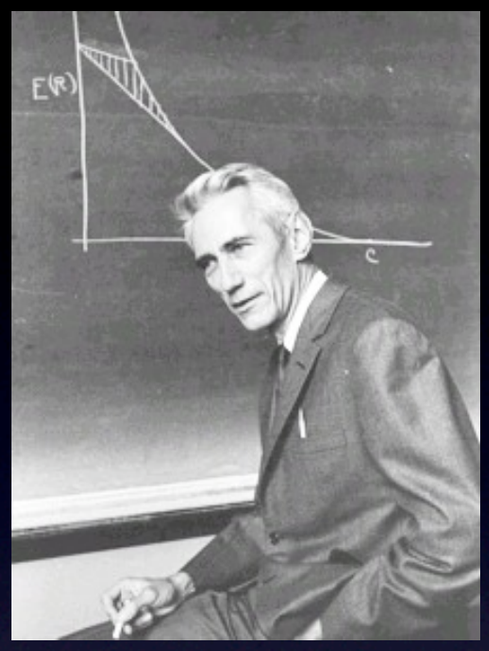
Claude Shannon

How to communicate in the presence of noise?

"A mathematical theory of communication" (1948)

Error correcting codes

Noisy Probabilistic Bits



Claude Shannon

How to communicate in the presence of noise?

"A mathematical theory of communication" (1948)

Error correcting codes



John von Neumann

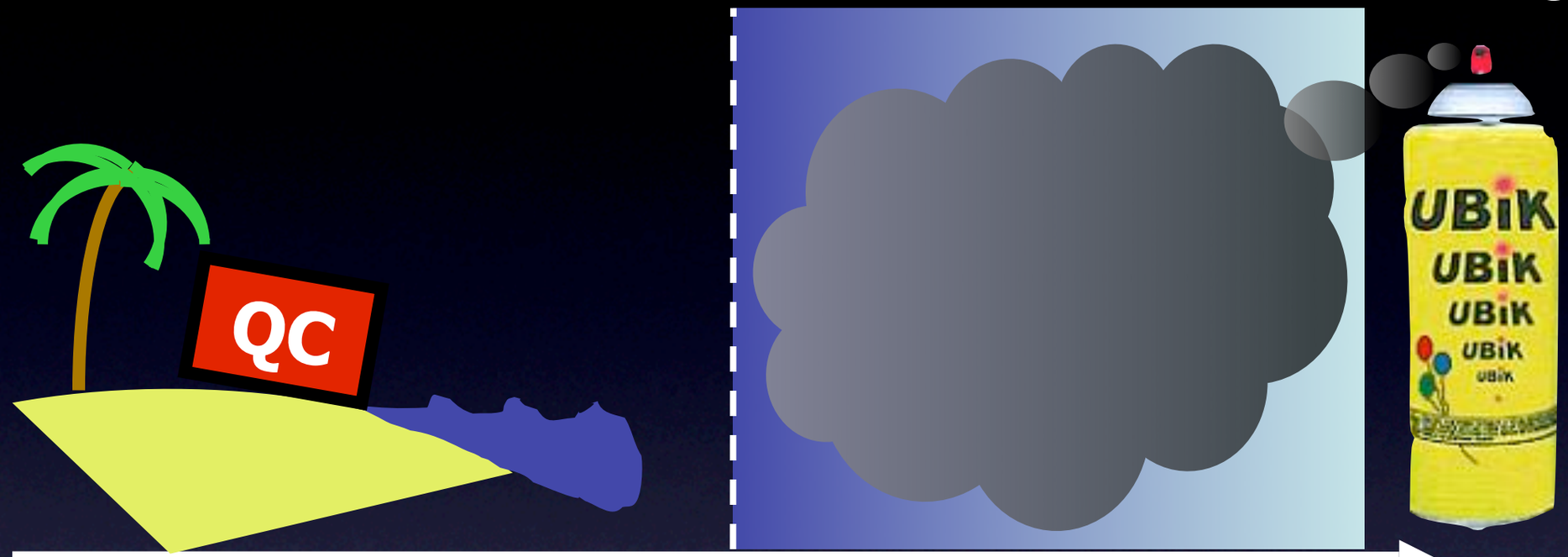
How to compute when the subsystems of your computer can fail?

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)

Fault-tolerant architectures

Fact: Fault-Tolerance

errosol



“decoherence”

“imprecision”

Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Potential: Quantum Hardware

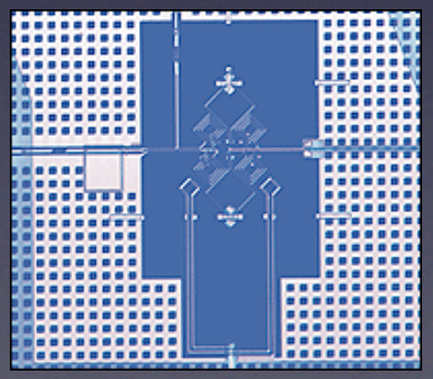
Solid State

superconducting circuits

electron spin in Phosphorus doped Silicon

quantum dots

defects in diamonds

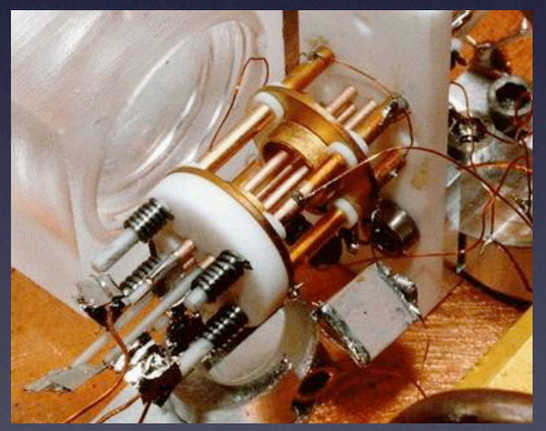


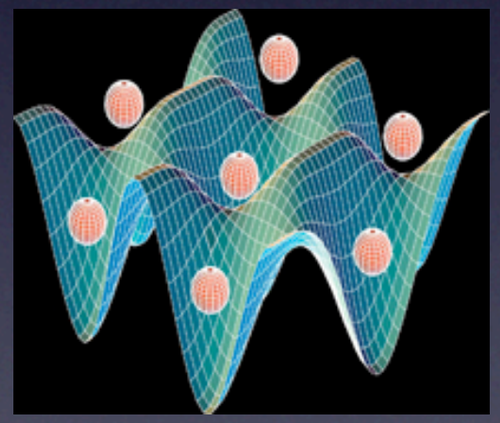
Atomic

cavity QED

neutral atoms in optical lattices

ion traps





Photon Based

linear optics + single photon devices

The Quantum Conundrum

Quantum computers can efficiently factor.

The Quantum Conundrum

Quantum computers can efficiently factor.

1. There exists a classical efficient algorithm for factoring!

The Quantum Conundrum

Quantum computers can efficiently factor.

1. There exists a classical efficient algorithm for factoring!

2. Quantum computers cannot be built

The Quantum Conundrum

Quantum computers can efficiently factor.

1. There exists a classical efficient algorithm for factoring!

2. Quantum computers cannot be built
   A. Quantum theory is wrong!

The Quantum Conundrum

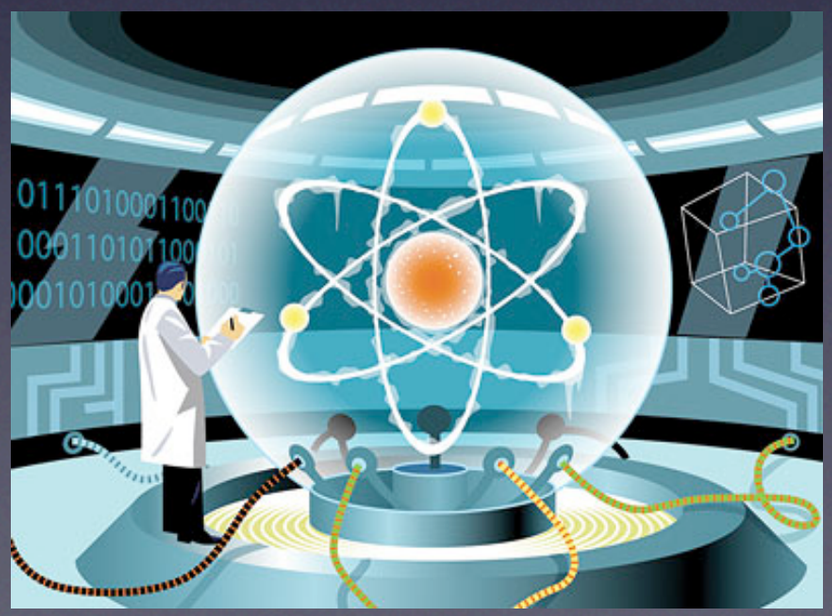
Quantum computers can efficiently factor.

1. There exists a classical efficient algorithm for factoring!

2. Quantum computers cannot be built
   A. Quantum theory is wrong!
   B. Some fundamental physical mechanism exists preventing quantum computers!

Quantum Computing Myth, Fact, and Potential

Dave Bacon
Department of Computer Science & Engineering
Department of Physics
University of Washington



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